CHAPTER 1: ELECTROLYSIS AND ELECTRICAL ENERGY

Electrolysis is defined as the chemical decomposition of an ionic compound (electrolyte) in fused or aqueous state
During the electrolysis of acidified water using platinum electrodes, water decomposes according to the balanced
In aqueous copper(II) sulphate solution, CuSO4(aq) <=> Cu^{2+}(aq) + SO4^{2-}(aq), where positively charged C

Table 1.1: Comparison of Strong and Weak Electrolytes

Electrolyte	Degree of Dissociation	Major Conducting Particles	Electrical Conductivity
Dilute H2SO4(aq)	Complete (nearly 100%)	Free mobile ions (H+ and SO4^{2-})	High conductivity
Acetic Acid CH3COOH(aq)	Partial (< 5%)	Mostly unionized molecules and few ions	Low conductivity

Ohm's Law states that the potential difference V across the ends of a metallic conductor is directly proportional to
Joule's Law of Heating states that the heat energy H produced in a resistor of resistance R carrying current I for ti
Fig. 1.1: Hoffman Voltameter apparatus for electrolysis of acidified water (Labels: Anode, Cathode, Graduated col
Coordinate Ion Check: $CuSO_4$ dissociates into Cu^{2+} and SO_4^{2-}

CHAPTER 2: THE FIRST WAR OF INDEPENDENCE (1857) AND INDIAN MONSOON

The Doctrine of Lapse was an annexation policy applied by Lord Dalhousie, which stated that if an Indian ruler of
The immediate cause of the Revolt of 1857 was the introduction of the Enfield rifle, whose cartridges were grease
First, on 29 March 1857 at Barrackpore, Mangal Pandey defied British officers; second, on 10 May 1857, sepoys
Whereas the Early Nationalists relied on constitutional petitions, prayers, and memoranda, in contrast the Assertiv

Table 2.1: Comparison of Arabian Sea Branch and Bay of Bengal Branch
Monsoon Branch	Primary Windward Barrier	Relative Moisture Content	Major Rainfall Region
Arabian Sea Branch	Western Ghats	Very high moisture	Western Coastal Plains and Mumbai
Bay of Bengal Branch	Garo-Khasi-Jaintia Hills	Moderate-high channeled moisture	Mawsynram, Cherrapunji,

Map 2.1: Isohyetal Distribution of Annual Rainfall Across the Indian Subcontinent
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CHAPTER 3: DIFFERENTIAL CALCULUS AND CELLULAR RESPIRATION

The Fundamental Power Rule of Differentiation states that for any real exponent n, the derivative of f(x) = x^n with
Aerobic cellular respiration is the catabolic process by which glucose is completely oxidized in the presence of oxy
The Bohr Effect refers to the physiological phenomenon whereby an increase in carbon dioxide partial pressure (p
Consequently, actively respiring tissues with elevated CO2 and lower pH trigger enhanced unloading of O2 from o